



F1-13-MKII / F1-13a Flow Over Weir

The Flow Over Weirs consists of five basic elements used in conjunction with the flow channel in the moulded benchtop of the Hydraulics Bench.

PRODUCT CODE: F1-13-MKII

Categories: Agricultural, Educational, F1 Series Fluid Mechanics, F1-13-MKII Flow Over Weir

Tags: Armfield , Characteristics Of Flow , Coefficient Of Discharge , F1-13-MKII , Flow Over Weirs , Fluid Mechanics , Hydraulics Bench , Rectangular Notch , Smooth Flow , Stiffing Baffle , V Notch , Vernier Hook And Point Guage

DESCRIPTION

The Flow Over Weirs accessory consists of five basic elements, used in conjunction with the flow channel, in the moulded bench top of the Hydraulics Bench. Two weir plates of different shapes are provided enabling familiarisation and comparison with theory.

A quick-release connector in the base of the channel is unscrewed and a delivery nozzle screwed in its place.

A stilling baffle locates into slots in the walls of the channel. The combination of the inlet nozzle and stilling baffle promote smooth flow conditions in the channel.

A Vernier hook and point gauge is mounted on an instrument carrier, which is located on the side channels of the moulded top. The carrier may be moved along the channels to the required measurement position.

The rectangular notch weir or V notch weir to be tested is clamped to the weir carrier in the channel by thumb nuts. The stainless steel weir plates incorporate captive studs to aid assembly.

F1-13a Advanced Weirs available: See Accessories

TECHNICAL SPECIFICATIONS

Overall dimensions of weir plate

Thickness: 0.002m

Width: 0.23m

Height: 0.16m

Dimensions of rectangular notch

Thickness: 0.002m

Width: 0.03m

Height: 0.082m

Angle of V notch weir: 90° inclusive

Hook & point gauge range: 0-150mm

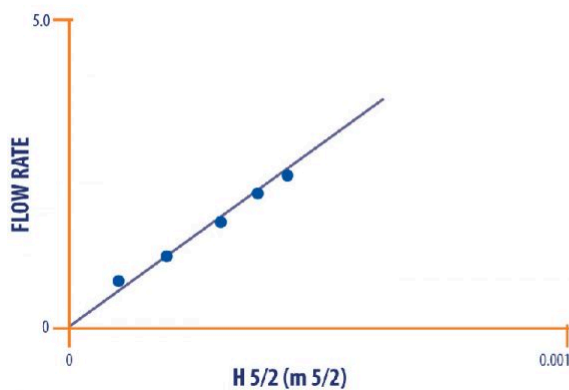
Accuracy: 0.1mm

FEATURES & BENEFITS

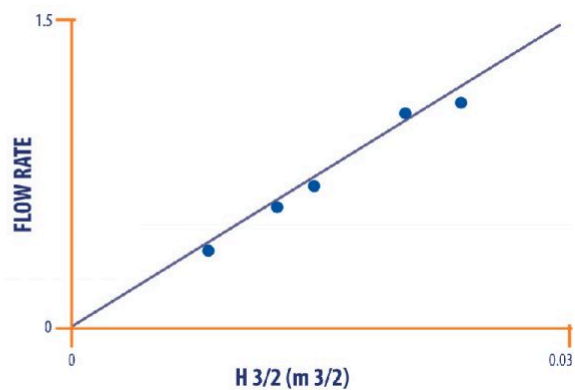
Demonstration Capabilities

- To determine the characteristics of open channel flow over a rectangular notch
- To determine the characteristics of open channel flow over a triangular (vee) notch
- To determine values of the discharge coefficient for both notches

(x10⁻⁴ m³/s)



(x10⁻³ m³/s)



Armfield – F1-13-MKII Flow Over Weirs Graph

Accessories

F1-13-MKII Weirs:

- Stiffing Baffle and Inlet Nozzle
- Vernier Hook & Point Gauge with Instrument Carrier
- Regular V Plate Weir 90°
- Large V Plate Weir 90°
- Large Notch Plate Weir 30mm
- Large Notch Plate Weir 50mm

F1-13a Advanced Weirs:

- Large Trapezoidal weir 30mm
- Large Trapezoidal weir 50mm
- Large V plate weir 30°
- Dog Bone weir



Armfield – F1-13a Advanced Weirs

Requirements

Hydraulics Bench Service unit F1-10

Dimensions

OVERALL DIMENSIONS OF WEIR PLATE

Thickness: 0.002m

Width: 0.23m

Height: 0.16m

DIMENSIONS OF RECTANGULAR NOTCH

Thickness: 0.002m

Width: 0.03m

Height: 0.082m

Angle of V notch weir: 90° inclusive



Hook & point gauge range: 0-150mm
Accuracy: 0.1mm
Requires Hydraulics Bench Service unit F1-10

Order Codes

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F1-13a

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